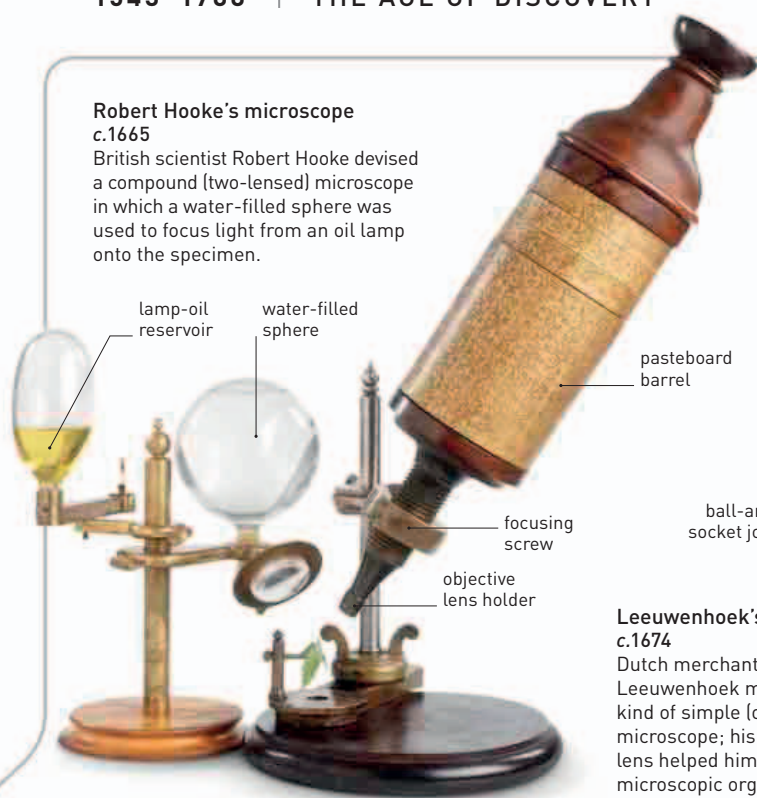
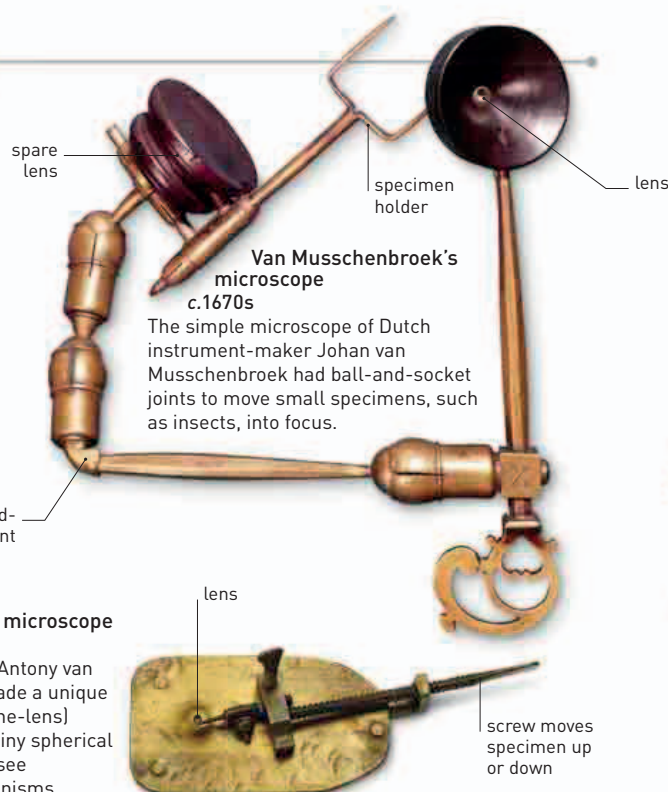


Robert Hooke's microscope
c.1665

British scientist Robert Hooke devised a compound (two-lensed) microscope in which a water-filled sphere was used to focus light from an oil lamp onto the specimen.

**Van Musschenbroek's microscope**
c.1670s

The simple microscope of Dutch instrument-maker Johan van Musschenbroek had ball-and-socket joints to move small specimens, such as insects, into focus.

**Leeuwenhoek's microscope**
c.1674

Dutch merchant Antony van Leeuwenhoek made a unique kind of simple (one-lens) microscope; his tiny spherical lens helped him see microscopic organisms.

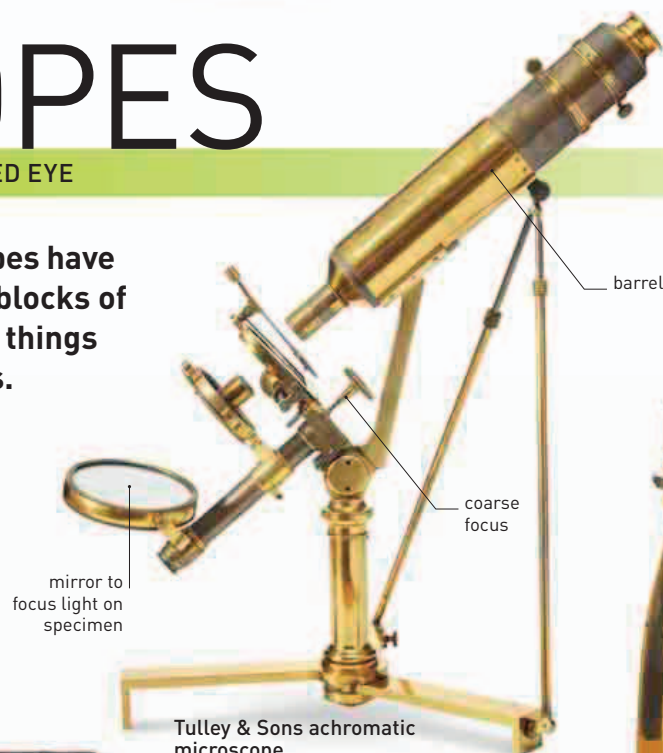


MICROSCOPES

VIEWING THE MICROSCOPIC WORLD BEYOND THE NAKED EYE

By opening up a world in miniature, microscopes have helped scientists to understand the building blocks of the world around us—from the cells of living things down to individual molecules and even atoms.

In the early 1600s, Dutch eyeglass-makers made the first microscopes by fixing two lenses together in a tube to create a magnifying power greater than that of a single lens used on its own. As lenses were refined, so the quality of the magnified image improved. Then, in the 20th century, breakthroughs in atomic physics led to the invention of the electron microscope, which—instead of light rays—used electron beams with shorter wavelengths to reveal even tinier particles.

**Tulley & Sons achromatic microscope**
c.1835

Designed by British scientist Joseph Lister, this microscope was made with new achromatic lenses that focused different colors accurately together, yielding better images.

instruction booklet

interchangeable objective lenses

Compound drum microscope
c.1850

The popular drum microscope focused on a specimen mounted on a basal stage, using a sliding body tube that contained the lenses. This design made it easy to transport the microscope and lenses.

**Culpeper compound microscope**
c.1740

British instrument-maker Edmund Culpeper produced inexpensive tripod-style microscopes; early models were made partly of wood. However, the fixed upright design and crude focus made them difficult and uncomfortable to use.

